1	$y = (2x+5)\mathrm{e}^{3x}$ $\frac{\mathrm{d}y}{\mathrm{d}x} = 2\mathrm{e}^{3x} + (2x+5)3\mathrm{e}^{3x}$ $\frac{\mathrm{d}y}{\mathrm{d}x} = 0 \Rightarrow 2+3(2x+5) = 0 \Rightarrow x = -\frac{17}{6}$	M1 A1 dM1A1 **(4)** **(4 marks)**

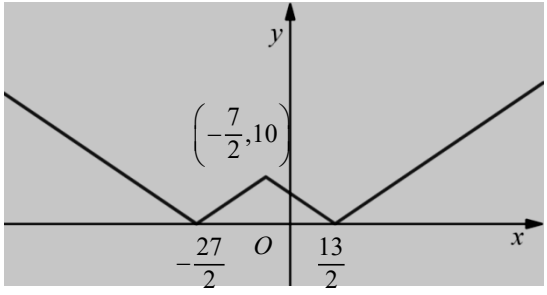
2 (a)	$8\cos\theta = 3\text{cosec}\,\theta$	
	States or uses $\text{cosec}\,\theta = \dfrac{1}{\sin\theta}$	B1
	Attempts to use $\sin 2\theta = 2\sin\theta\cos\theta$	M1
	$\sin 2\theta = \dfrac{3}{4}$ cso	A1
		(3)
(b)	Correct order of operations to find θ, Look for $\dfrac{\arcsin\left(\frac{3}{4}\right)}{2}$	M1
	$(\theta =)$ awrt $24.3°$	A1
		(2)
		(5 marks)

3 (i)	$\int (2x-5)^7 \, dx = \dfrac{(2x-5)^8}{16} + c$	M1 A1 **(2)**
(ii)	$\int \dfrac{4\sin x}{1+2\cos x} \, dx = -2\ln(1+2\cos x) \quad (+c)$	M1 A1
	$\int_0^{\frac{\pi}{3}} \dfrac{4\sin x}{1+2\cos x} \, dx = \left[-2\ln(1+2\cos x)\right]_0^{\frac{\pi}{3}} = -2\ln 2 + 2\ln 3 = \ln\dfrac{9}{4}$	dM1 A1 **(4)**
		(6 marks)

4	$$A = \frac{80pe^{0.15t}}{pe^{0.15t} + 4}$$	
(a)	$30 = \frac{80p}{p+4}$	M1
	$30p + 120 = 80p \Rightarrow p = \frac{120}{50} = 2.4$ *	A1*
		(2)
(b)	$50 = \frac{80 \times 2.4e^{0.15T}}{2.4e^{0.15T} + 4} \Rightarrow 72e^{0.15T} = 200$	M1 A1
	$\Rightarrow 0.15T = \ln\left(\frac{200}{72}\right) \Rightarrow T = 6.8$	dM1 A1
		(4)
(c)	80 m^2	B1
		(1)
		(7 marks)

5 (a)	Attempts to find y at -1.25 and -1.2 with one correct to 1sf	M1
	Achieves $y(-1.25) = -0.9$ and $y(-1.2) = 0.2$	
	With reason (change of sign and continuous) and Conclusion	A1
		(2)
(b)	(i) Attempts $\sqrt{12\ln(15)+8}$	M1
	$x_2 =$ awrt 6.3637	A1
	(ii) $x = 6.4142$	B1
		(3)
(c)	$\frac{dy}{dx} = \frac{12}{2x+3} - x$	M1 A1
	Stationary point when $\frac{12}{2x+3} = x \Rightarrow 2x^2 + 3x - 12 = 0 \Rightarrow x =$	dM1
	$\Rightarrow x = \frac{-3+\sqrt{105}}{4}$ or awrt 1.81 ONLY	A1
		(4)
		(9 marks)

6. (a)	$f(x)=\dfrac{5x-3}{x-4} \Rightarrow f'(x)=\dfrac{5(x-4)-(5x-3)}{(x-4)^2}=\dfrac{k}{(x-4)^2}$	M1 dM1
	States that $f'(x)=\dfrac{-17}{(x-4)^2} \Rightarrow f'(x)<0$ hence decreasing * cso	A1*
		(3)
(b)	$y=\dfrac{5x-3}{x-4} \Rightarrow xy-4y=5x-3 \Rightarrow xy-5x=4y-3$	M1
	$\Rightarrow x=\dfrac{4y-3}{y-5}$ So $f^{-1}(x)=\dfrac{4x-3}{x-5}$	A1
	Domain $x>5$	B1
		(3)
(c) (i)	$ff(x)=\dfrac{5\times\dfrac{5x-3}{x-4}-3}{\dfrac{5x-3}{x-4}-4}$	M1
	$ff(x)=\dfrac{5\times(5x-3)-3(x-4)}{5x-3-4(x-4)}=\dfrac{22x-3}{x+13}$	dM1 A1
(ii)	$5<ff(x)<22$	B1, B1
		(5)
		(11 marks)

7 (a)	$\left(-\frac{7}{2}, -10\right)$	B1 B1 **(2)**
(b)	Attempts to solve either $\frac{1}{2}(2x+7)-10 \ldots \frac{1}{3}x+1$ Or $-\frac{1}{2}(2x+7)-10 \ldots \frac{1}{3}x+1$	M1
	Both correct critical values $x \ldots -\frac{87}{8}, \frac{45}{4}$	A1
	Selects outside region for their critical values	dM1
	Correct solution $x \leqslant -\frac{87}{8}, x \geqslant \frac{45}{4}$	A1 **(4)**
(c)	 Shape Maximum One correct minimum Fully correct	B1 B1 ft B1 B1 **(4)** **(10 marks)**

8 (a)	$\log_{10} x = 2.74 - 0.079t$	
	$\log_{10} x = 2.74 - 0.079t$ $x = 10^{2.74} \times 10^{-0.079t}$ $x = 10^{2.74} \times \left(10^{0.079}\right)^{-t}$ | $x = p\,q^{-t}$ $\log_{10} x = \log_{10} pq^{-t}$ $\log_{10} x = \log_{10} p - t\log_{10} q$	
	A correct equation for p or q. E.g. either $\log_{10} p = 2.74$ or $\log_{10} q = 0.079$	M1
	A correct value for either p or q. Either $p =$ awrt 550 or $q =$ awrt 1.2	A1
	A correct equation for both p and q. E.g both $p = 10^{2.74}$ and $q = 10^{0.079}$	dM1
	Both $p =$ awrt 550 and $q =$ awrt 1.2 **with** proof*	A1*
		(4)
(b)	"p" is the amount of antibiotic (in mg) in the patient's bloodstream at the start	B1
		(1)
(c)	$x = 400 \times 1.4^{-t} \Rightarrow \frac{dx}{dt} = -400\ln 1.4 \times 1.4^{-t}$	B1
	Substitutes $t = 5$ into $\frac{dx}{dt} =$ awrt -25	M1 A1
		(3)
		(8 marks)

9 (i)	$2\sec^2 x - 3\tan x = 2$	
	$2\left(1+\tan^2 x\right) - 3\tan x = 2$	M1
	$2\tan^2 x = 3\tan x \Rightarrow \tan x = \frac{3}{2}, \left(\tan x = 0\right)$	dM1
	$x = \text{awrt } 0.983, \pi$	A1, B1
		(4)
(ii)	$\frac{\sin 3\theta}{\sin\theta} - \frac{\cos 3\theta}{\cos\theta} \equiv \frac{\sin 3\theta\cos\theta - \cos 3\theta\sin\theta}{\sin\theta\cos\theta} \equiv \frac{\sin\left(3\theta-\theta\right)}{\sin\theta\cos\theta}$	M1
	$\equiv \frac{\sin\left(3\theta-\theta\right)}{\frac{1}{2}\sin 2\theta}$	dM1, A1
	$\equiv \frac{\cancel{\sin 2\theta}}{\frac{1}{2}\cancel{\sin 2\theta}} \equiv 2$ *	A1*
		(4)
		(8 marks)

10 (a)	$x = y\mathrm{e}^{2y} \Rightarrow \quad \dfrac{\mathrm{d}x}{\mathrm{d}y} = \mathrm{e}^{2y} + 2y\mathrm{e}^{2y}$	M1, A1
	$\Rightarrow \dfrac{\mathrm{d}x}{\mathrm{d}y} = \dfrac{x}{y} + 2x$	
	$\Rightarrow \dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{1}{\dfrac{x}{y} + 2x}$	dM1
	$\Rightarrow \dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{y}{x + 2xy} = \dfrac{y}{x(1+2y)}$*	A1*
		(4)
(b)	Deduces $y = -\dfrac{1}{2}$	B1
	Substitutes $y = -\dfrac{1}{2} \Rightarrow x = -\dfrac{1}{2\mathrm{e}}$ Range for k $-\dfrac{1}{2\mathrm{e}} < k < 0$	M1, A1
		(3)
		(7 marks)